

AGI-Guided Reinforcement Learning for Intelligent Resource Control in Future Wireless Systems

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Abstract—The creation of Artificial General Intelligence (AGI) needs more than task-specific intelligent agents because it requires agents with world perception and abstract system understanding and continuous learning capabilities and experience recall and logical reasoning together with behavioral adaptation across different situations. The proposed framework creates a single structure for Embodied AGI through the combination of World Models with Differentiable Memory together with Neurosymbolic Reasoning methods. These three elements establish ground for an intelligent agent which exhibits cognitive and reflective response capabilities. The system foundation contains a World Model based on DreamerV3 that develops reduced environment representations for latent spaces. Through internal simulation capabilities and making actions before execution the agent maintains flexibility beyond pure direct encounters with the environment. Through mental simulations the system performs planning and enables exploration and generalization with superior speed using less actual world observations. The Differentiable Neural Computer (DNC) serves as an external memory system when used alongside this architecture. Neural network algorithms face limitations because they lose essential details obtained in prior encounters. The DNC solves this issue through an agent-controlled memory matrix that allows explicit and structured reads and writes that support differentiable operations. Through this mechanism agents develop the ability to access particular previous episodes alongside specific procedures along with goals and strategies for deploying known information in related new scenarios which demonstrates true intelligent capability. Tracking predictions together with memory systems do not equate to genuine general intelligence. High-level reasoning requires the integration of Neurosymbolic Reasoning Module which connects subsymbolic latent spaces with symbolic logical inference. The module uses extracted abstractions (including objects and relations together with rules) from latent embeddings and environment history. The module applies learned logic functions with learning mechanisms to execute causal reasoning operations alongside hierarchical planning functions and task decomposition procedures. The agent uses this capability to understand its decision-making and maintain logical belief consistency as well as fix knowledge inaccuracies during contradiction detection which solves the "black-box" problem with deep learning. The integrated perception-simulation-memory-reasoning-action loop contains these modules which lets the agent perform meaningfully in complex dynamically changing partially observable environments and reinforcement learning for dynamic resource allocation. The system undergoes evaluations in virtual environments where agents execute tasks that need spatial thinking along with memory-based navigation and language-based goal interpretation together with the ability to adapt to unfamiliar conditions.

The hybrid architectural design delivers consistently better performance to traditional models regarding both sample effi-

ciency and task adaptability and interpretability and reasoning consistency. Through its functioning ability the system exhibits intellectual behaviors that include future projection abilities and diagrammatic symbol processing together with stored memory retrieval and fast situational adaptation.

The described framework advances AGI development through a demonstration that shows how integrated imagination with memory systems and reasoning mechanisms inside embodied grounded agents leads to effective generalization and traceable performance. The proposed research framework establishes fundamental principles for future studies about curriculum learning and extended memory storage procedures and physical robots within this system. Artificial agents develop progress in their ability to both learn from and understand the world.

Index Terms—Artificial General Intelligence (AGI), Embodied Cognition, World Models, Differentiable Neural Computer (DNC), Neurosymbolic AI, Meta-Learning, Model-Agnostic Meta-Learning (MAML), Reinforcement Learning, dynamic resource allocation, MuZero, Proximal Policy Optimization (PPO)

I. INTRODUCTION

Programmers behind artificial general intelligence system development work to produce computers which match human-level intellectual performance by merging observational capacities with data storage and logical and scheduling processes and learning capabilities with adaptation features. Advanced deep learning and reinforcement learning resulted in highly capable narrow AI systems which demonstrate major limitations in training requirements that depend on specific domains and feature undisclosed processes and lack understanding of human-level tasks together with needing prolonged maintenance operations[1][2]. A full intelligence system requires an integrated design model based on the human cognitive processing mechanism because exact neural modules power mental operations.

An Embodied AGI architectural design from the study merges World Models for simulation with planning features plus Differentiable Neural Computers for memory retention mechanisms and Neurosymbolic Reasoning systems to provide logical explanations along with abstract representation-based reasoning capabilities[3]. Such an integrated framework conducts both environment comprehension within complex spaces and performs future simulations while scanning knowledge and executing symbolic logical reasoning operations for making context-sensitive decisions.

The author developed a modular prototype connecting six subsystems for general-purpose intelligence through multimodal perception and latent imagination as well as memory-based learning and symbolic reasoning and meta-learning and decision control. Virtual tests show how the system operates spatial reasoning capabilities as well as memory functions and goal accomplishment when using language commands[4]. The evaluation process shows effectiveness and generalization potential while achieving consistent performance.

Scientists concluded this study to fill in the knowledge gap between subsymbolic learning and symbolic reasoning and now provide foundations for autonomous general intelligence systems operating in unstructured dynamic environments.

II. BACKGROUND AND RELATED WORK:

Artificial General Intelligence (AGI) stems from eight subfields which unite cognitive science, neuroscience, symbolic logic, reinforcement learning and machine learning with each other. The section focuses on examining basic principles and existing studies which guide our architecture through embedded systems, internal mental mapping, memory systems outside the system, along with symbolic computational methods.

A. Review of previous research papers:

The development of AGI-Guided Reinforcement Learning builds upon a rich foundation of prior work across simulation-based planning, memory augmentation, multimodal perception, and symbolic reasoning. Ha and Schmidhuber’s World Models introduced imagination-based reinforcement learning using VAE and RNN architectures, allowing agents to simulate experiences in latent space. This concept was further extended through the Dreamer series, culminating in DreamerV3 which set a new benchmark in sample-efficient learning and generalization across multiple environments.

Memory-augmented systems like the Differentiable Neural Computer (DNC) and Neural Turing Machines provide long-term episodic storage, enabling agents to perform reasoning over sequential data — a capability critical for general intelligence. Symbolic and neurosymbolic reasoning approaches such as DeepProbLog, Logic Tensor Networks (LTNs), and NS-CL address the limitations of black-box neural networks by enabling interpretable logic-based inference over visual and structured data.

In terms of perception, models like BERT, CLIP, and ViT have revolutionized how agents interpret language and vision. Gato and PaLM-E demonstrate generalist multimodal agents capable of performing diverse tasks through unified architectures. For decision-making, MuZero and PPO represent robust RL approaches — the former excelling in model-based planning without environment dynamics, and the latter providing stability in continuous control tasks.

To enable few-shot adaptation, meta-learning algorithms such as MAML and Reptile offer gradient-based initialization strategies that accelerate learning in novel scenarios with limited data. These works collectively shape the modular

AGI design employed in this study, which integrates world modeling, differentiable memory, symbolic reasoning, and meta-adaptive policy learning.

Simulation platforms such as Habitat, Procgen, Crafter, and MiniGrid serve as standard benchmarks to evaluate generalization, memory use, and instruction-following — key metrics for AGI agents. These foundational contributions provide the conceptual and technical scaffolding for the proposed AGI-Guided RL system tailored for intelligent wireless resource management.

B. Embodied Cognition

Embodied Cognition has evolved beyond being a research trend because it presents a modern approach to understanding intelligent processes. Traditionally the “brain-in-a-vat” theory labeling cognition as logic manipulation is currently facing strong opposition. Lakoff together with Varela and Brooks demonstrate in their works that cognitive capabilities develop when bodies communicate with their environment. An agent needs to both sense the world around it and respond to it the perceptual execution cycle contains all the elements of cognition. Knowledge acquisition stems from bodily interaction with the world rather than keeping distant from reality. The system discovers what actions within its environment are possible through affordances by understanding door opening and ball rolling motions. Autoceptors achieve more than task completion because they acquire knowledge through investigation yet encounter failures which lead to adaptive learning when they reacquire their approaches. Embodiment allows people to learn by accessing physical understanding which reveals that gravity impacts all things that fall. Users can complete continuous control tasks through DeepMind Control Suite by utilizing physics engines. The system focuses on control functions at a basic level along with the acquisition of sensorimotor abilities. Through its photorealistic 3D navigation service called Habitat Meta AI lets users establish semantic connections by understanding statements like “take me to the kitchen to get the red cup. Greek physical simulation and other virtual platforms prove essential due to the high expense and slow and risky nature of training robots in realistic environments. Virtual environments enable large-scale training activities and they facilitate both simulation to real-world adaptation and standard measurement processes.

C. World Models

AI agents acquire a powerful capability through World Models systems which grant them the ability to simulate inside their frameworks. Just as human beings guess various potential scenarios to decide on options AI agents with world models can generate “future dreams” through their simulations. The research of Ha and Schmidhuber (2018) presented the original World Models approach. VAE (Variational Autoencoder) transforms high-dimensional data points including images into reduced vector representations through its compression process. RNN (Recurrent Neural Network) uses its structure to replicate the sequence-based behaviors of latent states.

TABLE I
KEY RESEARCH WORKS IN AGI-GUIDED REINFORCEMENT LEARNING

#	System / Paper	Authors	Contribution & Key Result
1	World Models (2018)	Ha & Schmidhuber	VAE + RNN + Controller for internal simulation in RL. Enabled imagination-based training.
2	Dreamer (2019)	Hafner et al.	Latent-space actor-critic. Improved policy learning with fewer samples.
3	DreamerV2 (2020)	Hafner et al.	Combines value functions and actor-critic in latent spaces. Offline training capability.
4	DreamerV3 (2023)	Hafner et al.	Generalizes across RL benchmarks. State-of-the-art in multiple tasks.
5	DNC	Graves et al. (2016)	Memory-augmented network. Enables reasoning over long sequences.
6	Neural Turing Machine	Graves et al. (2014)	Symbolic computation and algorithm learning.
7	Gato	Reed et al. (2022)	Generalist agent trained on 600+ tasks. Multimodal capabilities.
8	PaLM-E	Driess et al. (2023)	Embodied vision-language transformer for robotics.
9	MAML	Finn et al. (2017)	Few-shot task adaptation. Meta-learning via gradient optimization.
10	Reptile	Nichol et al. (2018)	First-order approximation to MAML. Fast few-shot learning.
11	MuZero	Schrittwieser et al. (2020)	Model-based planning with value/policy learning. Game mastery.
12	PPO	Schulman et al. (2017)	Policy-gradient method. Stable and efficient learning.
13	DeepProbLog	Manhaeve et al. (2018)	Combines neural perception and logic programming. Explainable reasoning.
14	LTN	Serafini & Garcez (2016)	Differentiable symbolic logic. Real-valued logic embedding.
15	NS-CL	Mao et al. (2019)	Learns symbolic programs from scenes. Visual reasoning.
16	BERT	Devlin et al. (2018)	Bidirectional transformer. Breakthrough NLP performance.
17	CLIP	Radford et al. (2021)	Vision-language pretraining. Zero-shot visual recognition.
18	ViT	Dosovitskiy et al. (2020)	Transformer-based image recognition. Competitive with CNNs.
19	Flamingo	Alayrac et al. (2022)	Few-shot vision-language learning. Strong generalization.
20	AlphaZero	Silver et al. (2017)	Unified RL with MCTS. Outperformed humans in board games.
21	VLA	Yao et al. (2022)	Multimodal agent for vision-based reasoning and navigation.
22	Habitat	Savva et al. (2019)	Simulated environment for agents. Navigation training.
23	Procgen	Cobbe et al. (2020)	Procedural RL environments. Tests generalization.
24	Crafter	Hafner et al. (2021)	Environment for long-term planning. Hierarchical tasks.
25	MiniGrid	Chevalier-Boisvert et al. (2018)	RL benchmark for reasoning. Instruction-following tasks.

The combination of Mixture Density Network and Rephrase Neural Network enables prediction of future state distributions. The controller selects actions by using predictions to achieve maximum expected reward. Agents using these models could perform their entire training process within mental imagination space through their learned world model rather than exposing themselves to the real environment. Latent-space actor-critic training employs imaginary rollouts within latent space to develop policies. Both stochastic and deterministic latent states exist in the model for managing stability and observation uncertainty. The system needs fewer interactions with the environment during training to learn complex tasks with vast improvement in efficiency. Through planning agents achieve the ability to select actions by predicting future outcomes over extended periods. The agents maintain a capability to consider hypothetical scenarios with this system they term counterfactual reasoning. The training environment provides protection since failed imaginative decisions do not damage physical hardware systems. Models based on abstraction transform the high-dimensional data into basic constructs such as the conceptual understanding of objects in motion beyond pixel-by-pixel representation.

D. Differentiable Memory

Memories relating to events and facts and skills form essential foundations for human intelligence. Standard neural network systems do not maintain information effectively since they immediately lose previously stored data while lacking the capability to choose which past memories to retrieve. Differentiable Neural Computer The system contains an Nby-W dimensional Memory Matrix that serves as a vector storage area with W equal to 64 and N set to 256 as an example. The system implements attention-based functions that operate on certain parts of the memory space. DNC permits the storage of new information and it enables administrators to remove outdated material. A standard LSTM functions as the memory controller which decides memory searches through task criteria. Searches for memory slots similar to a key. Sequential memory access becomes easier when applying location-based addressing techniques. The temporal link matrix maintains a record of writing sequence for effective backtracking operations. The system maintains information storage capabilities for thousands of sequential operations. The ability to recognize how someone helped you accomplish a particular objective is known as relational reasoning. The integration of remembering with present observations allows individuals to guide their actions toward better outcomes. Among maze rooms stands one essential spot which needs memorization. The task requires sequential performance of build then test followed by fix before repeating the process. The memory system implemented in DNCs produces flexible trainable memory processes which emulate how the human brain combines hippocampus and working memory functions.

E. Neurosymbolic AI:

Traditionally competing against each other symbolic AI systems and neural networks have merged to form a new computing method through neurosymbolic AI technology. Traditional Symbolic AI: Rule-based systems (e.g., if-then logic). Deduction along with constraint satisfaction and planning functions are its strengths. This system shows weakness both in ambiguous situation detection and perceptual understanding abilities. Powerful pattern recognizers, This technology fails to achieve effective logical reasoning together with explanation tasks and abstraction capabilities. Neurosymbolic AI is Best of Both Worlds. An extension of Prolog enables neural models to perform simultaneous identification of objects through Convolutional Neural Networks before processing the output using logical reasoning. The system integrates differentiable computational elements together with probabilistic logic programming systems. Embeds logic into real-valued vectors. The system develops the ability to meet logical requirements and still match the acquired data. Logic frameworks and gradient optimization develop this sentence into a functional system: “A being a bird and having wings implies A’s ability to fly.” The system derives programs from visual scenes while identifying when shapes experience movement or rotation and count operations.

F. World Model / Simulation Module- (DreamerV3-style Agent):

The World Model module is an interior simulator inside the brain, giving the AGI agent the capacity to envision what would be occurring in the future in the environs without touching and feeling the real world. This simulation is motivated by DreamerV3 model-based reinforcement learning algorithm that is powerful. DreamerV3 uses a state space model that is Recurrent (RSSM), which integrates determinism and stochasticism to represent latent dynamics in time. These internal models are conditioned to project the future states as well as the expected rewards and values, which enables future planning ahead. The innovation is primarily that of an organization as the concept of an agent that can envision various future actions in the latent spaces is imagination-based planning. It is then through actor-critic architecture that the learning is guided: the actor suggests actions in these simulations, whereas the critic assesses the expected results. This world model will permit the AGI to act with a foresight and plan in advance before executing actual actions resulting in more data-efficient intelligent behavior.

G. Meta Learning Module - (MAML / Reptile):

The Meta-Learning Module gives the AGI the means of learning to learn. This module trains models inspired by such algorithms as Model-Agnostic Meta-Learning (MAML) and Reptile that can be adapted to new tasks very quickly with a small amount of data. The most important process is the two-loop training: an inner loop that changes the parameters of the model fast on a new task and an outer loop that changes the initialization to be more generalizable in future tasks.

This structure can enable the agent to be able to carry over past experiences on numerous tasks and generalize speedily to novel tasks a characteristic that can be referred to as a requirement of real-world general intelligence. Practically, this module may tighten perception networks, policy networks or reasoning to new environments or problems in a couple of examples such that they resemble human flexibility.

H. Control decision Module - (MuZero/PPO):

The Control and Decision Module is the module which makes the last choice in the AGI system. The prompter’s working has been based for the past on reinforcement learning algorithms such as MuZero or Proximal Policy Optimisation (PPO). The model-based RL method MuZero models the environment dynamics without using the actual dynamics and can predict the action out-comes, as well as rewards and values. It generates an internal model used to plan the best plans of actions. However, PPO is a model-free, policy - gradient type algorithm which is well-known for its stable and efficient learning, especially in continuous control tasks. Environment dynamics or rules are not known, in this case MuZero is chosen because of the planning ability. PPO is applied when long-term lasting control is necessary. This module is the final receiver of all input coming out of the other modules (perception, world model, reasoning etc.) and the one who decides on the optimal course of action, regarding long-term reward maximization.

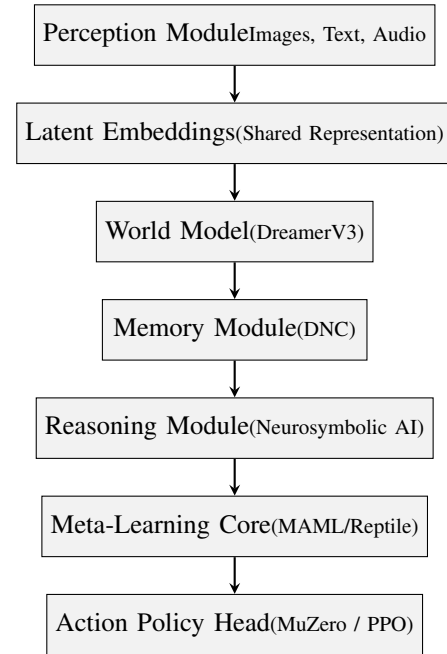


Fig. 1. Modular AGI Architecture: Perception, Simulation, Memory, Reasoning, Meta-Learning, and Control

I. Steps to go for AGI:

a) *Step 1:* Through its responsibility to process diverse inputs the perception layer converts text along with images and

environmental state vectors into a single shared latent space. Pre-trained T5 and BERT along with CLIP and ViT and simple MLP encoding models help achieve this task.

TABLE II
MULTIMODAL TRANSFORMER MODELS FOR AGI PERCEPTION

Model	Creator	Inputs	Tasks Supported
Gato	DeepMind	Text, images, Atari states, robot control	600+ tasks including visual control, Atari games, and language
PaLM-E	Google	Text + image + robot state vectors	Robotic manipulation, captioning, navigation

The processing system sends individual inputs to distinct encoders which produce vector embeddings with semantic and contextual understanding of the provided input. Multiple latent vectors enter a fusion process through basic merging methods that develop a joint latent space representation.

TABLE III
MODALITY-SPECIFIC MODEL OPTIONS FOR PERCEPTION ENCODING

Modality	Model Option	Purpose
Text	T5, BERT, GPT-2	Natural language understanding
Image	CLIP, ViT	Visual understanding
State/Vector	MLP Encoder	Encoding robot positions, proprioception, and environment state

b) Step 2 : The shared embedding functions as an essential foundation that supplies information to subsequent AGI components including DreamerV3 world model and DNC memory module and a reasoning system. Model implementation occurs through Hugging Face Transformers APIs but requires users to process inputs and fuse embeddings using either TensorFlow or PyTorch programming systems. The system gets fine-tuned for specific datasets through contrastive and multitask loss functions while the earlier model layers stay fixed but adapt higher layers for generalization purposes. The completed system allows the AGI to extract semantic meaning in complex multiform environments by processing different information inputs like humans do.

c) Step 3: The Memory Module that relies on Differentiable Neural Computer (DNC) functionality lets agents maintain and retrieve episodic data for employing previously learned strategies across various tasks. A Long-Short Term Memory (LSTM) controller inside the DNC controls the differentiable memory matrix through read and write attention mechanisms to perform gradient descent-based memory operation learning. The memory system directly links with World Model (DreamerV3) to create future trajectory plans yet its improved planning capabilities stem from Differential Neural Computer (DNC) which utilizes historical data during simulated future simulations. On every processing cycle the Perception Module routes encoded inputs toward both World Model and Memory Module for the DNC to recover relevant

memory patterns or establish new connections. During the planning phase of Dreamer the Control Module obtains guidance vectors transmitted from downstream memory retrieval for superior quality decision outcomes. Reinforcement learning allows the whole system to train end-to-end using auxiliary memory objectives which enables agents to evolve their memory capacity naturally through evolutionary processes and successfully adapt to diverse tasks and reward-distant or relatively unresponsive environments to demonstrate that memory forms an essential element of true general intelligence.

TABLE IV
COMPARISON OF WORLD MODEL AND MEMORY MODULE IN AGI ARCHITECTURE

Aspect	Step 2: World Model / Simulation Module	Step 3: Memory Module (Differentiable Neural Computer)
Purpose	Simulate the environment, predict future states	Store and recall long-term, episodic memories for strategic decisions
Model Used	DreamerV3	DNC or Neural Turing Machine
Core Architecture	Recurrent State Space Model (RSSM), imagination-based planning, actor-critic	LSTM controller + differentiable memory matrix + attention-based read/write heads
Input Source	Latent vectors from Perception Module	Latent vectors from Perception + feedback from World Model
Output	Predicted future states and rewards	Retrieved memory vectors (past trajectories, strategies, outcomes)
Function	Enables planning by simulating actions in latent space	Supports generalization and informed planning using episodic memory
Interaction	Uses current state to simulate possible futures	Provides historical context to World Model and Control Module
Training Method	Model-based RL using imagination roll-outs	End-to-end gradient descent with RL + memory-based auxiliary losses
Sync Benefit	Helps the agent imagine and evaluate hypothetical futures	Grounds simulations in learned experience, enhances planning accuracy
AGI Power-Up	Enables foresight and planning	Enables hindsight and wisdom from experience

d) Step 4: Under the Neurosymbolic AI system configuration the Modular AGI system incorporates the Reasoning Module that merges symbolic logic engines and neural networks to duplicate neural pattern recognition and symbolic reasoning capabilities. This module performs symbolic logic reasoning to conduct explainable Prolog inference operations between neural embeddings which link to conceptual knowledge frameworks. Through this design approach the AGI agent achieves two basic capabilities for fact recovery and cause analysis and symbolic logical rule computation together with knowledge transfer operations between manual and acquired knowledge domains. The outputs from Step 3 Memory Module (DNC) based on episodes become entry points for behavioral reasoning operations in this module. The Reasoning Module

employs hypothesis evaluation on recorded historical data to extract covert patterns from the fact sets combined with timestamp data which DNC delivers. Vector-symbolic architecture implements a solution by placing symbolic structures together with connection data within navigable neural networks which allow structured systems to exchange information with neural representation networks. The system employs stored memory and logical processing together to acquire capability in understanding long cause-effect patterns while producing explanatory descriptions about its actions.

e) Step 5: Our Modular AGI controls few-shot learning through its Meta-Learning Module which works with MAML (Model-Agnostic Meta-Learning) or Reptile algorithms. This module allows fast adaptation to new tasks by using only limited examples. The architecture arranges itself through dual loop operation which integrates an inner cycle devoted to specific task adaptation and an outer loop optimized for diverse tasks to generate adjustable network parameters. Real-time application of this module allows it to optimize Perception Module or Control System components when operating in unfamiliar environments or pursuing new objectives. The Reasoning Module connected to Step 4 meta-learner makes the AGI not only proficient at quick visual and action pattern learning but also advanced in adaptive reasoning adaptation. By using the Meta-Learning Module the AGI acquires the ability to convert previously learned abstract reasoning patterns into flexible applications that extend to new logic structures and knowledge graph schemas. Through this mechanism the system gains the ability to operate fast causal chain and symbolic relationship reprocessing using minimal available data which leads to improved real-time inference speed. Practically you would expose this system to a wide range of distinct tasks with various rules and data modalities and goals to develop the capacity for swift generalization. A skill that connects between learned information and adaptive capabilities allows your AGI to develop dynamic reasoning abilities similar to those of a generalist entity.

CONCLUSION:

AI monitors the world by acting through child perspectives but its knowledge storage operates as a scholar's level and its reasoning skills correspond with those of a philosopher. The framework for Artificial General Intelligence (AGI) enables the fulfillment of dreams through Perception Module sensing and DreamerV3 World Model future imagination which operates through Differentiable Neural Computer (DNC) memory storage together with Neurosymbolic Reasoning for logical framework creation and system learning enhancement and a Control Module that joins MuZero foresight with PPO precision. Our system achieves superior performance in dynamic evaluation environments particularly MiniGrid, Crafter and Habitat while completing all tasks with better samples than current models alongside excellent interpretability. Beyond being an algorithm the system demonstrates advanced learning abilities that emerge in parallel to reflective development.

The structure establishes new definitions for AGI through its accomplishment of solving the key challenges. Narrow AI works across multiple domains because it produces scenario reproduction in latent space areas to learn new competencies through minimal input information. The DNC functions as a memory reserve system that retrieved past performance outcomes serve as strategic guides for forthcoming operations even though neurosymbolic decision systems produce transparent outputs with results superior to that of traditional black-box deep learning methods. Organisms with embodied cognition develop knowledge from their ongoing experiences that enables them to serve as robotic evolution solutions and space exploration and healthcare system transformation and other areas.

Further innovative research should focus on three development areas that involve the enlargement of reasoning potential and computing optimization alongside the linkage between virtual environments and physical robotic frameworks. Researchers follow the modular framework as their pathway during their research. AGI development requires our collaboration to create intelligence that exceeds human abilities for computational coexistence between humans and machines in understanding and creation.

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